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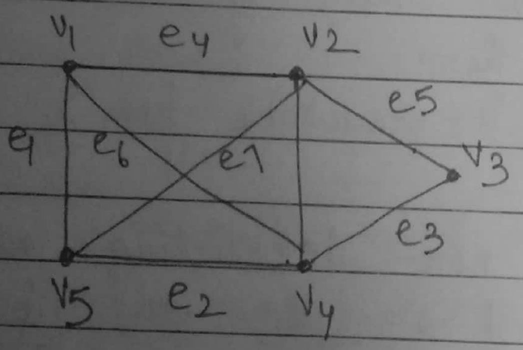
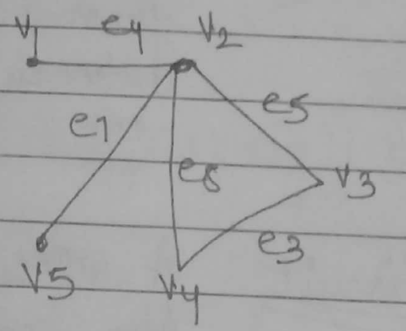
More terms related to Graphs →

1) **Walk** An alternating sequence of vertices and edges starting and ending in vertices such that every edge between 2 vertices in the sequence is incident on the 2 vertices.

(i) The first and last vertex of a walk - Terminal Vertices
(ii) A walk that begins and ends on the same vertex - closed walk
(iii) A walk that has different terminal vertices - open walk.

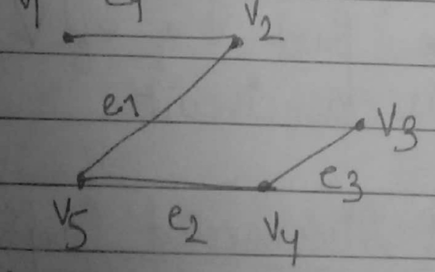


Graph



[Traversing all vertices]

walk1 $v_1 e_4 v_2 e_8 v_4 e_3 v_3 e_5 v_2 e_7 v_5$



walk2 $v_1 e_4 v_2 e_7 v_5 e_2 v_4 e_3 v_3$

*) In a walk, vertex can appear more than once, but an edge cannot appear more than once.

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2) <u>Path</u> An open non-intersecting walk i.e
(All vertices in the walk are distinct and no vertex appears more than once)
Also called simple path, or, elementary path.

(*) Terminal vertices have degree 1 and other vertices have degree 2.

eg Walk 2 is a path
walk 1 is not a path

(i) <u>Length of a path</u> → Number of edges in the path

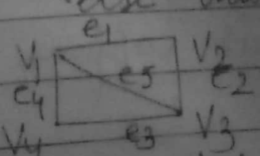
(ii) <u>Reachable</u> → Two vertices are reachable from each other, if there exists a path between them.
eg. V_1 is reachable from V_4

3) <u>Circuit</u> A closed non-intersecting walk.
Also called a <u>Cycle</u> or <u>Circular Path</u>

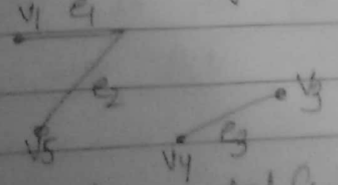
*) the degree of every vertex is 2.
eg the part $V_2\ e_8\ V_4\ e_3\ V_3\ e_5\ V_2$ of walk 1 is a circuit.

4) <u>Connected & Disconnected Graph</u> A Graph is connected if every two vertices are reachable from each other else disconnected.

eg.



a) Connected Graph



b) Disconnected Graph

Every disconnected graph can be partitioned into connected subgraphs called Components

eg (b) has 2 components.

Theorem A disconnected simple graph G (without self loops and parallel edges) with n vertices and k components can have atmost

$$\frac{(n-k)(n-k+1)}{2} \text{ edges.}$$

5) Homomorphism & Isomorphism

Isomorphism ($G \cong H$) An isomorphism from graph G to a graph H is a bijective mapping

$$\alpha : V(G) \rightarrow V(H)$$

such that,

$$(x,y) \in E(G) \iff (\alpha(x), \alpha(y)) \in E(H)$$

(α maps edges to edges
α maps non edges to non edges)

$\therefore$ Both adjacency and non adjacency is preserved.

Homomorphism A homomorphism from a Graph G to a Graph H is a mapping (not necessarily bijective)

$$\alpha : V(G) \rightarrow V(H)$$

Such that,

$$(x,y) \in E(G) \Rightarrow (\alpha(x), \alpha(y)) \in E(H)$$

α maps edges to edges

But α may map a non edge to
- single vertex
- edge
- non edge

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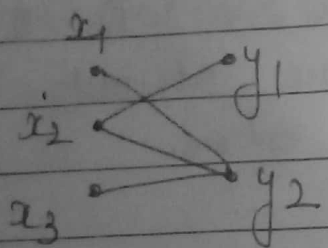
*) An isomorphism would be a homomorphism but in general, the vice-versa may not be true.

<u>Example</u> A bipartite graph G has a homomorphism to K_2.



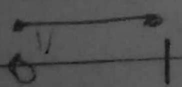
Let $V(G) = X \cup Y$ where X & Y are the partite sets of G

Thus every edge $e \in E(G)$ is of the form $e(x,y)$ where $x \in X$ & $y \in Y$

α:

Let $V(K_2) = \{0,1\}$

Define $\alpha : V(G) \to V(K_2)$ by



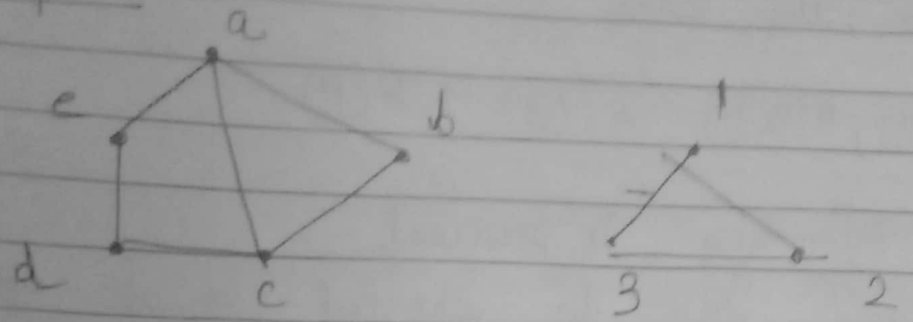
$$\alpha(v) = \begin{cases} 0 & \text{if } v \in X \\ 1 & \text{if } v \in Y \end{cases}$$

Now any edge $(x,y) \in E(G)$ has $x \in X$ and $y \in Y$ has $(\alpha(x), \alpha(y)) = (0,1) \in E(K_2)$

$\therefore$ α is a homomorphism from G to K_2.

*) In this example non-edges are either mapped to a single vertex eg $(x_1, x_2) \to 0$ or to an edge eg $(x_1, y_1) \to (0,1)$

Example 2



Consider $\alpha : V(G) \to V(H)$ given by

$\alpha(a) = 1$
$\alpha(b) = 2$
$\alpha(c) = 3$
$\alpha(d) = 2$
$e(e) = 3$

Check edges

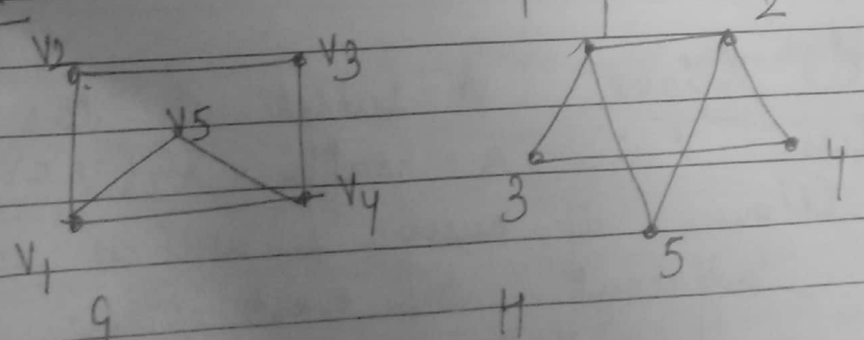
$(a, b) \to (1, 2)$
$(c, d) \to (3, 2)$

$\therefore$ α is a homomorphism from G to H

Non-edge $(e, b) \to (3, 2)$
" " $(b \to d) \to$ single vertex 2

Example 3

Show Isomorphism



Let $\theta : V(G) \to V(H)$ given by

$$\theta : \begin{pmatrix} V_1 & V_2 & V_3 & V_4 & V_5 \\ 1 & 3 & 4 & 2 & 5 \end{pmatrix}$$

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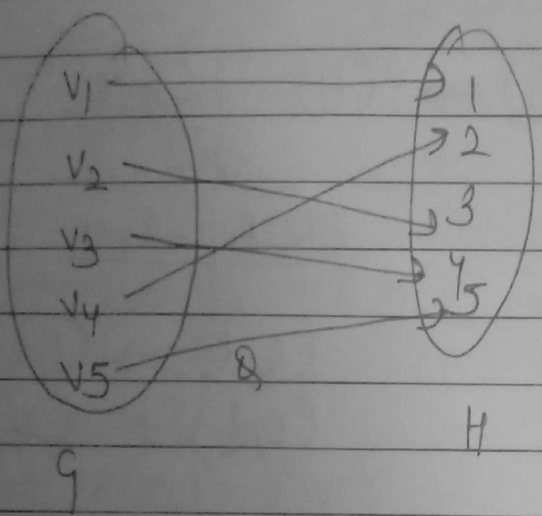
Consider the edge $(v_1, v_2) \in E(G)$

$(\theta(v_1), \theta(v_2)) = (1,3) \in E(H)$

$\therefore$ Adjacency is preserved

The same can be checked for all other edges.



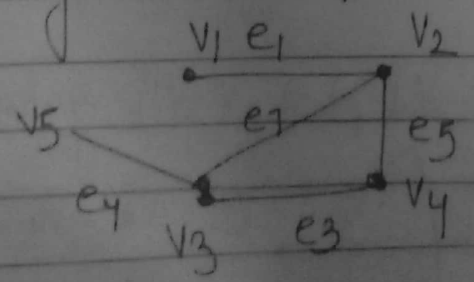
(1) one - one & (2) onto

$\therefore$ It is a bijection

*) All non- edges of G map to a non-edge in H in case of ~~Homom~~ Isomorphism.

6) <u>Chromatic Number</u>

(i) <u>Clique</u> A subset of the set of vertices V in a Graph G(V, E) is called a Clique, if every 2 vertices in the set are adjacent to each other.



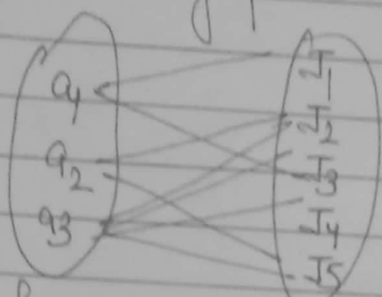
The subsets $\{v_1, v_2\}$, $\{v_2, v_3\}$, $\{v_2, v_4\}$, $\{v_3, v_5\}$, $\{v_2, v_3, v_4\}$ are Cliques

Matching Let us consider 2 disjoint set of vertices V_1 and V_2. V_1 is a set of 3 vertices and each vertex in V_1 represents an applicant.

V_2 is a set of 5 vertices and each vertex represents a job available.

An applicant is eligible for more than 1 job as shown in the graph below →

eg1



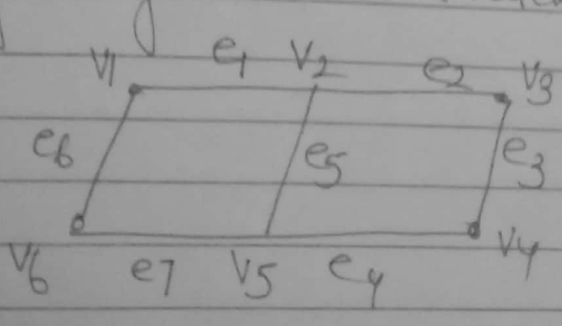
$\{(a_1 J_1), (a_2 J_2), a_3, J_4)\}$ is a Matching

The problem of assigning each applicant a job for which he or she is eligible is the problem of matching one set of vertices to another set of vertices.

A matching in a graph is a subset of edges in which no 2 edges are adjacent.

A single edge is also a matching

eg2



The sets $\{e_1 e_7\}$, $\{(e_2, e_4)\}$, $\{e_3, e_5, e_6\}^c$ are some matchings.

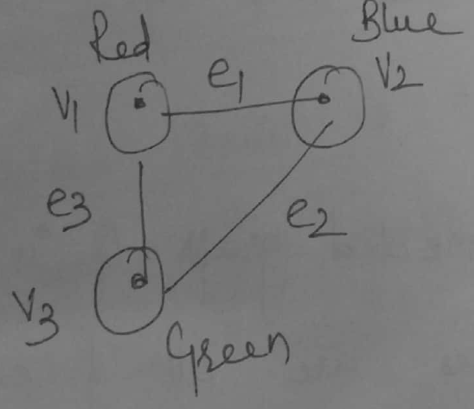
Chromatic Number →

(i) Colouring of a Graph : Assigning colours to the elements (vertices, edges, regions) of the graph such that no 2 adjacent elements have the same colours.

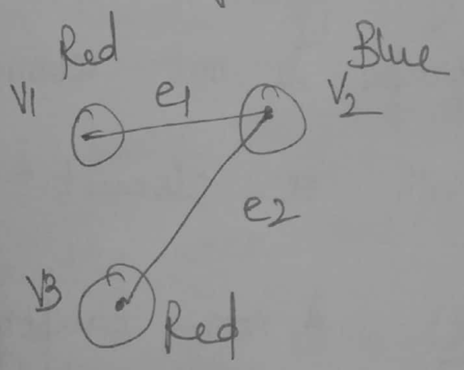
(ii) Proper Colouring : Assigning colours to all vertices of a graph such that no 2 vertices have the same color.

Such a graph is called a properly coloured graph.

(iii) Chromatic Number $K(G)$: is the minimum number of colours required to colour a graph properly.



3- Chromatic Graph



2- Chromatic Graph

*) Euler Line & Euler Graph →

*) A closed walk that contains all edges of a Graph is called a Euler line

*) Euler Graph : A Graph that contains Euler line

1) A Euler Graph is always Connected.
2) Euler Graph does not have isolated vertices. [closed walk]

*) Euler Path This Path traverses each edge in G exactly once. [open walk]

*) Euler Graph contains Euler Circuit.
*) If any value is zero for degree of vertices, then graph is not connected and does not have a Euler Path or circuit.

Theorem 1 A non-empty connected graph G is Eulerian if and only if, its vertices are all of even degree.

Theorem 2 A connected graph contains a Euler path, but not a Euler Circuit if and only if, it has exactly two vertices of odd degree.

*) A Graph is either Euler Graph or has an Euler Path.

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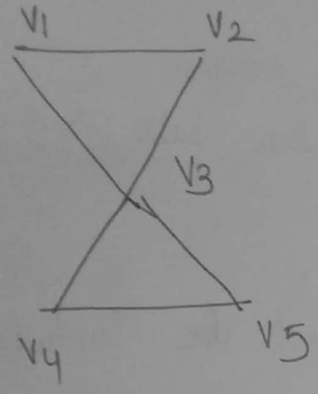
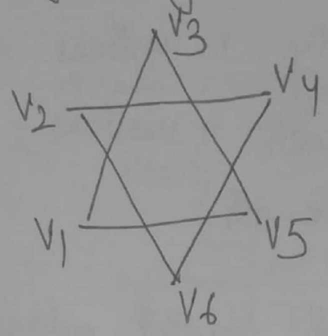
Q Check whether the following have Euler Circuits.





1) Yes

2) No (Not Connected)

(iii) No (Odd Degree)

*) Hamiltonian Path & Hamiltonian Circuit →

*) Hamiltonian Circuit → A closed walk that traverses each vertex of a graph G exactly once except the starting vertex at which the walk also terminates.

*) Hamiltonian Path → A path obtained by removing one edge from a Hamiltonian Circuit.

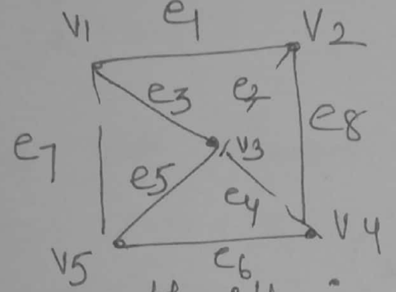
$$HC \Rightarrow HP$$
$$HP \not\Rightarrow HC$$

*) In a graph with n vertices, hamiltonian circuit contains exactly n vertices and n edges.

*) In a graph with n vertices, hamiltonian path contains n−1 edges (length)

*) Self loops and parallel edges cannot be included in a hamiltonian circuit (path)

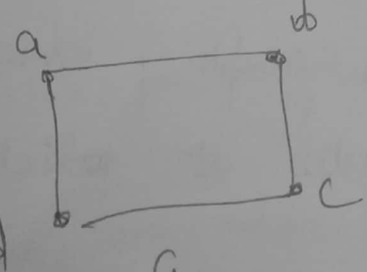
eg.



Hamiltonian Circuit

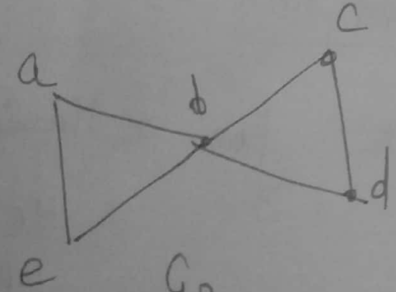
Q Give an example of a graph which contains →

(i) A Eulerian Circuit that is also a Hamiltonian Circuit



G_1

(ii) A Eulerian Circuit but not a Hamiltonian Circuit.

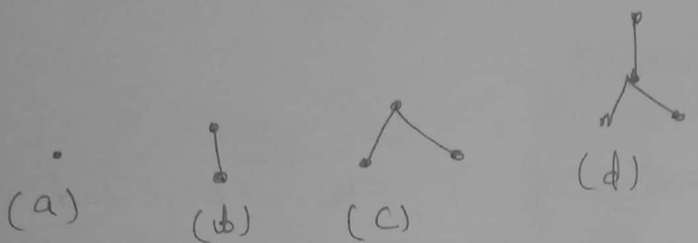


G_3

Trees

Trees A connected graph without any circuit (as subgraph) is called a Tree.

Forest A disjoint union of trees.



Trees with different vertices

$n=1, n=2, n=3, n=4$

A tree is a simple connected graph without any self loops and parallel edges.

Theorem 1 There is one and only one path between every pair of vertices in a tree T.

Theorem 2 There are $n-1$ edges in a tree with n vertices.

∴ A Tree is an undirected graph T that satisfies any of the following equivalent conditions →

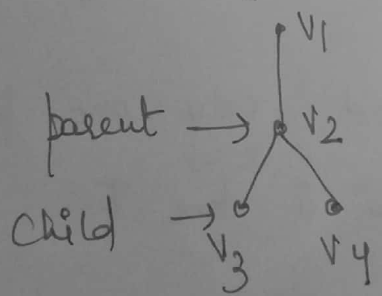
1) T is connected and has no circuits.

2) T has no circuits and a circuit is formed if any edge is added to T.

3) T becomes disconnected if any edge is removed from T.

1) Every 2 vertices of T are connected by a unique path.

*) <u>Rooted Tree</u> If one vertex is designated as the root in the tree and its is distinguishable from other vertices.

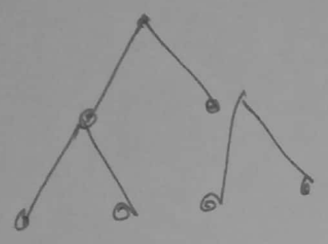


Rooted tree with root v_1

→ Every vertex except root has a unique parent.

→ A leaf is a vertex without children.

*) <u>Binary Tree</u> If each vertex has atmost 2 children (0,1,2 can be the # of children)



*) <u>Height of Binary Tree</u> A vertex v in a binary tree T is said to be at level l, if its distance from the root is l.

The height of a binary tree is the max level of any vertex in a tree.

Eg.